

IT Help Desk System

Requirements, Scope, and Technology Plan

Project phase	Analysis and Design
Frontend	React (JavaScript), HTML5, CSS3
Backend	PHP Laravel
Database	MySQL 9.7 (managed with DBeaver)
Repository	github.com/bashirsaad99/IT-Helpdesk-System

1. Project overview

The IT Help Desk and Ticketing Management System is a web application for managing technical support requests inside a company. Employees submit problems, IT agents investigate and resolve them, and administrators manage users, assignments, categories, departments, and reports.

2. Objectives

- Centralize technical-support requests in one traceable system.
- Give employees clear ticket status and communication history.
- Help IT agents prioritize, investigate, and resolve assigned work.
- Give administrators operational control and reporting visibility.

3. Scope

In scope	Out of scope for Phase 1
Authentication and role-based access	Live chat and video support
Ticket creation, assignment, status, priority	AI-based ticket classification
Comments, solutions, and attachments	Mobile applications
User, role, category, department management	External customer billing
Dashboards and basic reports	Advanced SLA automation

4. User roles

Role	Responsibilities
Employee	Create tickets, track status, add comments, upload attachments, and close resolved tickets.
IT Agent	View assigned tickets, update status, request information, communicate with employees, and add solutions.
Administrator	Manage users, roles, departments, categories, ticket assignments, and reports.

5. Functional requirements

Authentication and authorization

The system shall authenticate users securely and restrict screens and actions according to Employee, IT Agent, and Administrator roles.

Ticket creation

An employee shall create a ticket with a title, category, description, priority, and optional attachments. The system shall assign a unique ID and an initial Open status.

Ticket assignment and processing

An administrator shall assign a ticket to an IT agent. The agent shall move it through Assigned, In Progress, and Resolved states and record a solution.

Communication

Employees and IT agents shall add timestamped comments. When more information is requested, the employee can reply and the agent can review the ticket again.

Ticket closure

After a solution is received, the employee or an authorized administrator shall close the ticket. Closed tickets remain available for history and reporting.

Administration and reports

Administrators shall manage users, roles, departments, and categories and shall view ticket counts and status summaries.

6. Business rules

- Every user has exactly one role; a department is optional.
- Every ticket has one requester and one category; an assignee is optional until assignment.
- Only IT agents and administrators may change processing status or record a solution.
- A ticket may have many comments and attachments.
- Deleting a ticket deletes its comments and attachments; users referenced by ticket history are protected.
- Valid priorities are Low, Medium, High, and Urgent.
- Valid statuses are Open, Assigned, In Progress, Resolved, and Closed.

7. Non-functional requirements

Area	Requirement
Security	Password hashing, server-side validation, role checks, protected file uploads, and HTTPS in deployment.
Usability	Responsive, consistent interfaces with clear status labels, errors, and confirmations.
Performance	Common dashboard and ticket operations should respond quickly under normal company load.
Reliability	Database constraints and controlled status transitions shall protect data integrity.
Maintainability	React frontend, Laravel API, and MySQL database shall be separated into clear layers.
Auditability	Ticket timestamps and conversation history shall show what happened and when.